



SIMATS ENGINEERING

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Course Code : CSA0204

Slot: A

Course Name :C-PROGRAMMING

Course Faculty: DR.A.SEVALKURAM

Final Assignment Title: SMART CAMPUS ENERGY MONITORING AND ENERGY-SAVING RECOMMENDATION SYSTEM USING C

Module Photographs:

```
C smart_campus.c X
C: > Users > sridh > Downloads > C smart_campus.c
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAX_FACILITIES 10
5 #define DAYS 7
6
7
8 typedef struct {
9     char name[40];
10    float limit;
11    float energy[DAYS];
12    float total;
13    float average;
14    float savings;
15    float score;
16 } Facility;
17
18 void getInput(Facility f[], int n);
19
20 int main() {
21     Facility f[MAX_FACILITIES];
22     int n;
23     printf("\nSMART CAMPUS ENERGY MONITORING SYSTEM\n");
24     printf("Enter number of facilities (1-10): ", MAX_FACILITIES);
25     scanf("%d", &n);
26
27     // PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
28     PS C:\Users\sridh\Downloads> gcc smart_campus.c -o smart_campus
29     PS C:\Users\sridh\Downloads> .\smart_campus.exe
30
31     SMART CAMPUS ENERGY MONITORING SYSTEM
32     Enter number of facilities (1-10): 2
33
34     Enter facility 1 name: Library
35     Enter energy limit (kwh) for Library: 500
36     Enter energy used for day 1: 60
37     Enter energy used for day 2: 55
38     Enter energy used for day 3: 70
39     Enter energy used for day 4: 65
40     Enter energy used for day 5: 50
41     Enter energy used for day 6: 45
42     Enter energy used for day 7: 55
43
44     Enter facility 2 name: Computer Lab
45     Enter energy limit (kwh) for Computer Lab: 800
46     Enter energy used for day 1: 120
47     Enter energy used for day 2: 110
48     Enter energy used for day 3: 130
49     Enter energy used for day 4: 125
50     Enter energy used for day 5: 115
51     Enter energy used for day 6: 105
52     Enter energy used for day 7: 120
53
54     ===== ENERGY REPORT =====
55     Facility    Limit(kwh)    Total(kwh)    Average(kwh)    Savings(kwh)    Score(%)
56     Library      500.00      400.00      57.14      100.00      80.00
57     Computer Lab  800.00      825.00      117.86     -25.00      0.00
58     =====
59     Total Campus Energy Used : 1225.00 kwh
60     Overall Average per Facility per Day : 87.50 kwh
61     Program completed successfully.
62     PS C:\Users\sridh\Downloads> []
```

Project Description:

My contribution to the Smart Campus Energy Monitoring and Energy-Saving Recommendation System was focused on understanding the problem, designing the C program structure, implementing important modules, and validating the system. I contributed to the facility data structure used to store the facility name, permitted energy limit, seven-day consumption values, total consumption, average consumption, savings and sustainability score.

I worked on the analysis logic used to calculate total and average energy consumption and identify the highest and lowest consumption days. I also contributed to the decision-making logic that classifies a facility according to its permitted energy limit and generates appropriate energy-saving recommendations. Functions and pointers were used to make the program modular and easier to maintain.

Another part of my contribution was testing the program with normal, boundary and invalid inputs. I checked negative energy values, invalid menu choices, facility searching, sorting based on consumption, reduction target validation and report-file creation. These tests helped verify correct behavior under different conditions.

This project improved my understanding of C programming concepts including structures, arrays, strings, functions, pointers, decision-making, loops, searching, sorting and file handling. It also demonstrated how programming can support responsible energy consumption by helping campus administrators identify excessive usage and make suitable energy-saving decisions.

Through this work, I gained practical experience in problem analysis, modular programming, debugging, validation and teamwork. Future improvements could include real-time sensor data, database connectivity, graphical dashboards and automated alerts for abnormal energy consumption.

Student Signature**Guide Signature**